

PREPARED FOR

T6N, R4W, Mount Diablo Meridian

COORDINATES 38.3459°N 122.3047°W WGS84	ELEVATION 66 ft NAVD88 · MSL	STATE PLANE CA 2 (0402) · N 1,887,922.81 E 6,474,289.05 (US ft, NAD83) -	FEMA ZONE Zone X 06055C0510FF	DECLINATION 13.0° E WMM 2026
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APN 038050009000 | ZONING AP | ACRES 0.481 ac

BEST GNSS WINDOW Mon, Jul 13 · 6:00 AM – 10:00 AM | PDOP 0.99 · 34 sv | NEAREST CORS
P261 · 14.1 mi ✓

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Site file · Google Earth
Download KMZ

RECORDER
Napa County Recorder-Clerk · 1127 First
St, Napa, CA 94559

NEAREST CORS
P261 · 14.1 mi

GENERATED
July 13, 2026

ASSESSOR
Napa County Assessor · 1127 First St,
Napa, CA 94559

NEAREST BM
JT9010 · 0.2 mi

SYSTEM
TopoScout · info@toposcout.com

This report compiles data from public and third-party sources (NGS, FEMA, USGS, NOAA, and county records) current as of the generation date shown above. GNSS planning values are predictive and conditions may change. This document is provided for planning purposes only and does not constitute a boundary survey, legal land description, or engineering certification. Verify all critical values against primary sources.

SITE & SIGNAL REPORT

COORDINATES

38.3459, -122.3047

WGS84 / NAD83

ELEVATION

66 ft

Above MSL (approx)

MAGNETIC DECLINATION

13.0° E

Compass reads 13.0° too far W

FEMA FLOOD ZONE

X

AREA OF MINIMAL FLOOD HAZARD · Panel 06055C0510FF

IONOSPHERIC CONDITIONS

PEAK KP INDEX

4.7

AVG KP

2.4

SOLAR FLUX F10.7

138 sfu

TEC (VTEC) ▲ DEGRADED

22.7 TECU

-1.8 TECU vs median

ASSESSMENT

MODERATE ACTIVITY

Peak Kp 4.7 (avg 2.4). Solar flux F10.7 = 138 sfu. Moderate geomagnetic activity. Some ionospheric delay possible. Static sessions should use longer occupation times.

GNSS / DOP SUMMARY

DATE	AVG PDOP	BEST 4-HR WINDOW	AVG SATS	RATING
Mon, Jul 13	0.99	6:00 AM – 10:00 AM	34	Ideal

GPS + GLONASS + Galileo + BeiDou MEO · Full multi-constellation model · 15° elevation mask · Per-day independent computation
GNSS satellite geometry is accurate within a 50-mile radius of the entered location – orbital mechanics do not change meaningfully at ground-level distances. Weather data is specific to the entered coordinates. For site-specific weather, terrain mask angle, and nearest CORS station, order an On-Demand Site Report.



WILDFIRE PROXIMITY

No Active Fires Within 150 Miles

No active fires as of report date (July 13, 2026) · Conditions may change – verify before fieldwork



GPS CONSTELLATION HEALTH

▲ 1 Satellite Currently Unhealthy – PRN 11

Active outage per NAVCEN · Affected satellites auto-excluded by receiver

Minor Obstruction

Low vegetation or sparse development nearby — minor sky obstruction possible.

Low Multipath

Minor reflective surfaces present — multipath generally manageable.

RECOMMENDED ELEVATION MASK

13° Minor vegetation — slight horizon reduction

✓ PDOP Check at 13°: 0.8–0.8 across survey window
PDOP remains ≤ 0.8 at 13° — geometry confirmed acceptable

5-POINT SAMPLE (CENTER + N/S/E/W · ~100 M)

Point	Cover Type	Obstruction	Multipath
Center	Dwarf Scrub	Low	Low
North	Dwarf Scrub	Low	Low
South	Dwarf Scrub	Low	Low
East	Dwarf Scrub	Low	Low
West	Dwarf Scrub	Low	Low

Dominant cover: Dwarf Scrub · Source: USGS NLCD 2024 via MRLC · 30 m resolution · Industry standard mask guidance per NGS field procedures

High 92°F | Low 60°F | Wind max 12 mph | Gusts 14 mph | Precip 0.00" | Sunrise 5:56 AM | Sunset 8:33 PM

Waning Crescent 2%

GNSS: Avg PDOP 0.99 · Best 4-hr window 6:00 AM – 10:00 AM (PDOP 0.94, 34 sats avg)

GNSS / DOP - HOURLY			
	PDOP	SATS	RATING
6:00 AM	0.84	36 sv	IDEAL
7:00 AM	0.95	34 sv	IDEAL
8:00 AM	0.95	33 sv	IDEAL
9:00 AM	1.00	33 sv	GOOD
10:00 AM	1.01	29 sv	GOOD
11:00 AM	1.11	25 sv	GOOD
12:00 PM	1.03	26 sv	GOOD
1:00 PM	1.08	26 sv	GOOD
2:00 PM	0.93	32 sv	IDEAL
3:00 PM	0.97	33 sv	IDEAL
4:00 PM	0.98	33 sv	IDEAL
5:00 PM	0.99	33 sv	IDEAL
6:00 PM	1.02	30 sv	GOOD
7:00 PM	0.99	32 sv	IDEAL



GO - GOOD CONDITIONS

Partly cloudy | Wind: 11.6 mph (gusts 14.1 mph) | Precip: 0" | Temp: 59.6°–92°F



DRONE WIND ANALYSIS



GO Peak: 11.6/14.1 mph S

* Sunrise: 5:56 AM | Sunset: 8:33 PM | fly window clipped to daylight

Best flight window: 6:00 AM – 9:00 PM (15 hrs)

Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≈turbulence	DIR
5 AM	1/3 mph		ESE
6 AM	1/2 mph		NW
7 AM	1/1 mph		WNW
8 AM	2/3 mph		SSE
9 AM	3/4 mph		S
10 AM	3/4 mph		SSE
11 AM	6/7 mph		S
12 PM	8/10 mph		S
1 PM	8/9 mph		S
2 PM	9/11 mph		S
3 PM	9/11 mph		S
4 PM	12/14 mph		S
5 PM	8/13 mph		SSW
6 PM	10/12 mph		S
7 PM	9/11 mph		SSE
8 PM	5/8 mph		SSE

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ▶ FREE TO FLY
Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107
→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY ↗](#)
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

FIELD PLANNING REPORT

N/A — NAPA COUNTY, CA (38.3459, -122.3047)

GENERATED: MONDAY, JULY 13, 2026

SURVEY DATE: MONDAY, JULY 13, 2026

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VERDICT: GO

Single clearest reason: PDOP geometry is excellent (avg 0.99, best window 0.94), weather is acceptable (overcast, no precipitation), and ionospheric activity is moderate with manageable delay. No constraints prevent field operations.

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BEST SURVEY WINDOW: 6:00 AM – 10:00 AM PDT

Window justification: Peak constellation strength with PDOP 0.94 and 34 satellites (multi-constellation). Morning timing avoids peak heat (forecast high 92°F) and provides stable atmospheric conditions ahead of afternoon development. This 4-hour span captures optimal geometry before midday degradation.

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BEST 4-HOUR STATIC GNSS WINDOW: 6:00 AM – 10:00 AM PDT

Recommended static occupation: Minimum 2 hours at primary marks within this window. Multi-constellation reception (GPS + GLONASS + Galileo + BeiDou) will mitigate the loss of GPS PRN 11 (currently unhealthy). Moderate ionospheric activity (Kp 4.7 peak, TEC 22.7 TECU slightly below median) warrants standard post-processing with IGS rapid products. No extended sessions required; standard static protocols apply.

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DRONE OPERATIONS ASSESSMENT: APPROVED WITH CAUTION

Wind forecast (max 12 mph) is within typical small UAS operating limits. Overcast conditions acceptable for visual line-of-sight operations. Confirm no TFR or airspace restrictions in Napa County. Operations can proceed in the recommended 6:00–10:00 AM window; afternoon thermal activity may increase turbulence.

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FIELD SAFETY WARNINGS

SATELLITE HEALTH: GPS PRN 11 is currently unhealthy (status: UNUSUFN). Receivers will auto-exclude. PDOP metrics already account for reduced constellation; no manual intervention required.

IONOSPHERIC DELAY: Moderate geomagnetic activity (Kp 4.7) may introduce $\pm 1\text{--}2$ cm additional atmospheric delay in static solutions. Use standard ionosphere-free linear combinations (L3/LC) and validate with rapid orbit/clock products post-mission.

WEATHER: Overcast skies pose no safety risk to GNSS operations but may affect drone visual operations and crew comfort. No precipitation forecast. Heat (high 92°F) is manageable with standard hydration and shade protocols.

SITE CONDITIONS: Dwarf scrub cover and terrain-based 13° elevation mask recommended. Multipath and sky obstruction risks are low. No active wildfires detected. Air quality is good (AQI 45).

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SUMMARY

Conditions are favorable for GNSS survey operations on Monday, July 13. The optimal survey window is 6:00–10:00 AM, when PDOP

reaches 0.94 with 34 satellites across all constellations. Moderate ionospheric activity and the loss of GPS PRN 11 do not materially degrade geometry. Weather is acceptable (overcast, 12 mph winds, no rain). Execute static sessions within the morning window using standard multi-constellation post-processing; drone operations approved with attention to wind and airspace verification. No field safety concerns preclude mobilization.

COUNTY RECORDING OFFICES

RECORDER / REGISTER OF DEEDS

Napa County Recorder-Clerk
1127 First St, Napa, CA 94559

ASSESSOR / TAX OFFICE

Napa County Assessor
1127 First St, Napa, CA 94559

CORS STATIONS – WITHIN 30 MILES

QR / LINK	STATION ID	NAME / LOCATION	DISTANCE
 NGS ↗	P261	HUNTERHILLCN2004	14.1 mi
 NGS ↗	P198	PETALUMAIRC2004	17.5 mi
 NGS ↗	OHLN	OHLONE PARK	23.5 mi
 NGS ↗	P196	MEACHUMLFLCN2006	24.0 mi
 NGS ↗	CASR	SANTA ROSA CA	24.9 mi
 NGS ↗	P267	DIXONAVIATCN2005	26.3 mi

NGS BENCHMARKS – WITHIN 10 MILES

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
 Datasheet ↗	JT9010	"13"	19.100m / 62.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9018	"13 E"	17.970m / 59.0ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9009	"13"	17.900m / 58.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9017	"13 D"	17.890m / 58.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9034	"20 A"	20.730m / 68.0ft NAVD88	Order	0.3 mi
 Datasheet ↗	JT9012	"13"	19.060m / 62.5ft NAVD88	Order	0.4 mi
 Datasheet ↗	JT9043	"25 A"	18.710m / 61.4ft NAVD88	Order	0.4 mi
	JT9016	"13 D"	18.200m / 59.7ft NAVD88	Order	0.5 mi

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
Datasheet					

FEMA FLOOD ZONE

X – AREA OF MINIMAL FLOOD HAZARD

Panel: 06055C0510FF

Effective: Sep 29, 2010

[View FEMA Map ↗](#)



DATA COLLECTOR IMPORT – NGS BENCHMARKS

NGS BENCHMARK CONTROL POINTS – Copy / Paste to Data Collector

Datum: NAD83 | Vertical: NAVD88 | Source: NGS NDE API

PID	NAME	LAT	LON	ELEV_FT	ORDER
JT9010	13	38.348333	-122.302778	62.7	
JT9018	13 E	38.348333	-122.302778	59.0	
JT9009	13	38.346667	-122.301111	58.7	
JT9017	13 D	38.345000	-122.301111	58.7	
JT9034	20 A	38.345000	-122.309444	68.0	
JT9012	13	38.351667	-122.304444	62.5	
JT9043	25 A	38.340000	-122.302778	61.4	
JT9016	13 D	38.341667	-122.297778	59.7	

[Download .txt File](#)

USGS LAKES & RESERVOIRS – within 150 miles

STATION	DIST	LEVEL	FULL POOL	LINK
CLEAR LK A LAKEPORT CA #11450000	58.1 mi	5.05 ft	N/R	USGS ↗

CALIFORNIA RESERVOIRS (DWR/CDEC) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
Lake Del Valle	60.6 mi	696.94 ft (-9 ft)	706 ft	36K af	67%
Folsom Lake	66.5 mi	448.22 ft (-18 ft)	466 ft	788K af	81%
Lake Oroville	93.4 mi	861.25 ft (-39 ft)	900 ft	2,859K af	81%
New Melones Lake	100.1 mi	1,020.22 ft (-68 ft)	1,088 ft	1,664K af	69%
Black Butte Lake	101.2 mi	—	228 ft	74K af	54%

USBR RESERVOIRS (Bureau of Reclamation) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
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ARMY CORPS RESERVOIRS (USACE) – within 150 miles

RESERVOIR	DIST	POOL ELEV	FULL POOL	STORAGE	% CAP
Del Valle	64 mi	696.9 ft	695.01 ft	—	100%
Camanche	89 mi	230.3 ft	221.86 ft	—	104%
Englebright Lake-Pool	95 mi	525.1 ft	522.35 ft	—	101%
Farmington Dam-Pool	99 mi	124.9 ft	117.09 ft	—	107%
Black Butte-Pool	101 mi	457 ft	458.36 ft	—	100%
Don Pedro	137 mi	—	N/R	—	—

USGS GROUNDWATER WELLS – within 15 miles

STATION	DIST	WELL DEPTH	WATER LEVEL	DATE	AQUIFER
005N004W32A002M	7 mi	200 ft	87.71 ft BGS	2014-11-05	N100CACSTL
005N004W28R001M	7 mi	51 ft	41.90 ft BGS	1977-10-03	N100CACSTL
004N004W04C001M	8 mi	80 ft	11.20 ft BGS	1977-10-03	N100CACSTL
004N004W05B001M	8 mi	—	54.10 ft BGS	1977-10-03	N100CACSTL
005N004W31R001M	8.1 mi	295 ft	—	—	—
004N004W05D002M	8.3 mi	60 ft	14.60 ft BGS	1977-10-03	N100CACSTL
004N004W06B001M	8.3 mi	201 ft	21.86 ft BGS	2022-06-07	—
004N004W02L001M	8.8 mi	—	9.90 ft BGS	1977-10-02	N100CACSTL

○ AI AQUIFER INTELLIGENCE — CLAUDE HAIKU

The N100CACSTL aquifer is the Napa-Sonoma Coastal Range alluvial and older sedimentary formation, which consists of Quaternary and Tertiary sand, gravel, silt, and clay deposited by ancient fluvial and marine processes. This layered geology creates confined to semi-confined conditions with variable permeability, meaning water moves slowly through clay-rich zones but more readily through sand and gravel lenses. At this site, water availability is moderate to good depending on which saturated layer a well penetrates, but sustained yield depends on local recharge and the degree of confinement.

FORMATION

Name	Napa-Sonoma Coastal Ranges alluvial and older sedimentary aquifer
Age / Rock	Quaternary to Tertiary age; dominantly sand, gravel, silt, and clay
Extent	400–600 sq mi (est. USGS/DWR)

DEPTH & RECHARGE

Depth Range	51–295 ft (5–30 floors)
Typical Depth	80–120 ft (8–12 floors)
Recharge Source	Precipitation on upland areas and lateral subsurface inflow from surrounding bedrock

Transit Time	20–50 years to 100 ft depth
WATER QUALITY	
pH	est. 6.8–7.4 (est. USGS/DWR)
Hardness	est. 180–350 mg/L CaCO3 (est. USGS/DWR)
TDS	est. 350–550 mg/L (est. USGS/DWR)
Notes	No sample data — typical for formation; moderate hardness and TDS expected for coastal range alluvial deposits
TREND	
Trend	Water levels show moderate long-term decline with significant fluctuation; 1977 measurements (9.90–54.10 ft BGS) versus 2014–2022 measurements (21.86–87.71 ft BGS) indicate drawdown in deeper zones
Source	USGS NWIS historical and recent field measurements
Seasonal Swing	8–20 ft estimated seasonal range based on well depth variance and formation characteristics

Source: USGS National Water Information System · Search all wells in this area → USGS NWIS Mapper ↗ · Water level data pending USGS API migration for CA state well numbers

LAND SUBSIDENCE & AQUIFER COMPACTION

✓ **LOW SUBSIDENCE RISK — No known critically overdrafted basin**
Source: CA DWR SGMA Bulletin 118 2024.1 · California only - full US pending server migration

🕒 AI SUBSIDENCE ASSESSMENT — CLAUDE HAIKU	
The Castro Valley area (coordinates 38.3459, -122.3047) is underlain by the North Coast Ranges aquifer system (N100CACSTL), which benefits from well-drained soils and is situated outside any critically overdrafted groundwater basin designated under SGMA. The geological setting—characterized by consolidated bedrock and adequate recharge from the Bay Area’s relatively robust precipitation patterns—has prevented the chronic groundwater over-withdrawal that typically triggers clay compaction and subsidence. Surveyors working in this area can proceed with standard elevation control procedures without the elevated scrutiny required in subsiding regions.	
SUBSIDENCE ASSESSMENT	
Susceptibility	Low
Primary Mechanism	None - consolidated terrain
Annual Rate	<0.5 mm/yr (est.)
Historic Loss	None documented
Aquifer Compaction	None - aquifer not over-pumped
SGMA Status	Not applicable - not in SGMA basin
BENCHMARK IMPLICATIONS	
Monument Risk	Low - monuments stable
Elevation Validity	Good - no active subsidence
GNSS Recommendation	Standard benchmark verification adequate
SURVEY ACTION	
Required Action	No special action required
Dataset	CA DWR SGMA Bulletin 118 2024.1
Coverage	California critically overdrafted basins only - full US coverage pending

STRUCTURAL & ENGINEERING PARAMETERS

SEISMIC (ASCE 7-22)		WIND / SNOW / ICE / FROST (ASCE 7-22)	
Site Class	D (default)	Basic Wind Speed	92 mph
Risk Category	II	Ground Snow Load	5.8 psf
		Winter Wind (W2)	0.35
		Ice Thickness	None
		Frost Depth	12 in minimum
		Est. from IECC Zone 3C — AFI data unavailable	
		Verify with local building official before construction	
		Source: ASCE 7-22 Hazard Tool · gis.asce.org	

CLIMATE ZONE (IECC 2021): Zone 3C · Marine · Marine - mild year-round, coastal influence (SF, LA coast) County: Napa

🕒 AI CLIMATE SUMMARY — RESIDENTIAL DESIGN	
This marine climate means your building envelope should prioritize air sealing and moderate insulation (R-13 to R-15 walls is sufficient), as you're managing consistent mild temperatures rather than extreme swings. Maximize south-facing glass for passive solar gain during cooler months and minimize west-facing glass to avoid overheating, while using overhangs or exterior shading on all glazing to manage the marine sun; north-facing glass can remain more open since heating load is modest. The key passive move here is natural ventilation through operable windows—design cross-ventilation paths to take advantage of consistent coastal breezes during your warm season, allowing you to reduce cooling loads significantly without relying on air conditioning.	
🕒 AI STRUCTURAL ASSESSMENT — CLAUDE HAIKU	
This coastal California site presents a well-drained gravelly loam foundation with moderate wind exposure and light snow loading, typical of marine microclimates. Seismic parameters are currently unavailable and may warrant immediate clarification with the local building official, as this region may fall within a seismically active area requiring detailed site-specific analysis. Standard spread footing foundations are anticipated to be feasible, though a geotechnical investigation is recommended to confirm soil bearing capacity and confirm Site Class D classification.	
FOUNDATION	
Type	Conventional spread footing anticipated

Basis Well-drained gravelly loam with favorable drainage characteristics suggests adequate bearing capacity; however, consult a licensed engineer to confirm allowable bearing pressure and verify no shallow groundwater complications

STRUCTURAL IMPLICATIONS

- Seismic** Seismic design category and spectral acceleration parameters not available — verification with local building official strongly recommended to determine whether SDC B, C, or D applies; seismic detailing may warrant special moment frames or intermediate braced frames pending confirmation
- Wind Exposure** Exposure C (open terrain with scattered obstructions) anticipated for ASCE 7-22 Risk Category II; verify site surroundings and topography to confirm exposure classification
- Wind** 92 mph design wind speed (3-second gust) anticipated — standard wind load design with typical connection detailing likely adequate; no hurricane-force exposure anticipated
- Snow** Ground snow load of 5.8 psf with winter wind factor W2=0.35 anticipated — standard roof design likely adequate for this light load; consult engineer to confirm roof slope and drift loading

INVESTIGATIONS & INSPECTIONS

- Geotech Required** Yes — geotechnical investigation recommended to confirm Site Class D, verify bearing capacity, assess liquefaction screening, and clarify seismic site response
- Liquefaction Screen** Screening recommended — Site Class D in marine coastal zone may warrant liquefaction assessment depending on groundwater depth; consult engineer and local building official
- Special Inspections** Anticipated requirements depend on seismic design category confirmation — may be required if SDC C or D applies; defer to engineer and building official
- Code Edition** ASCE 7-22 / IBC 2021 (planning reference only)
- Site Class Note** Site Class D assumed for planning purposes based on IECC estimate — geotechnical investigation strongly recommended to confirm soil profile, groundwater conditions, and seismic site class per ASCE 41 or local standards

SOILS — USDA SOIL SURVEY

- MAP UNIT:** Coombs gravelly loam, 2 to 5 percent slopes
- SERIES:** Coombs
- DRAINAGE CLASS:** Well drained
- HYDRO GROUP:** C
- SOIL ORDER:** Alfisols
- SUBORDER:** Xeralfs

🔍 AI SOIL INTELLIGENCE — CLAUDE HAIKU

Coombs gravelly loam is a well-drained Alfisol (Xeralf) formed in mixed alluvial and residual material on gently rolling terrain in the California Coast Ranges. The soil profile features a gravelly loam surface layer overlying a clay-enriched substratum, typical of Mediterranean climate soils with winter moisture and summer drought. Site development should account for moderate gravel content affecting excavation, potential clay migration into fill material, and the need for moisture conditioning during compaction.

Mixed alluvial and residual parent material on 2–5% slopes; characteristic of oak woodland and grassland vegetation in semi-arid Mediterranean climate.

DRAINAGE

- Class** Well drained
- Seasonal Saturation** None within 60 inches
- Flooding Risk** Minimal on 2–5% slopes
- Ponding Risk** None

ENGINEERING

- Bearing Capacity** 3,000–4,000 psf (shallow foundations)
- Rating** Good
- Shrink-Swell** Moderate — clay-enriched substratum
- Frost Action** Low
- Cut Suitability** Good (gravel-rich surface and subsurface material)
- Fill Suitability** Fair (clay substratum — moisture control and compaction verification required)

SEPTIC

- Suitability** Conditional
- Notes** Perc test required — clay loam substratum and gravel content may limit infiltration; seasonal water table absent but clay layer can restrict drainage in saturated conditions

HAZARDS

- Excavation** Moderate — 15–35% gravel content throughout profile; standard heavy equipment suitable
- Expansive Clay** Low — typical Alfisol clay mineralogy (not montmorillonite-dominant)
- Erosion** Low on slopes ≤5%; moderate risk in disturbed areas and concentrated flow zones; gravel mulch reduces surface erosion
- Other** None

Data sources: USGS National Water Information System · waterdata.usgs.gov · California DWR / CDEC · cdec.water.ca.gov · USBR RISE · data.usbr.gov · USACE CWMS · water.usace.army.mil · USGS NWIS Groundwater · waterservices.usgs.gov · USDA Web Soil Survey · sdmdataaccess.nrcs.usda.gov · CA DWR SGMA Bulletin 118 · gis.water.ca.gov · ASCE 7-22 Hazard Tool · gis.asce.org · USGS Design Maps · earthquake.usgs.gov · IECC 2021 Climate Zones · DOE Building America · Real-time data subject to revision

Water levels current as of report run: Jul 13, 2026, 12:51 AM UTC

* Live pool elevation not reported by this reservoir — full pool design elevation shown

OIL & GAS WELLS California — within 10mi radius

8 total
0 active
7 plugged
 By status:
 7 Plugged
 1 Idle

By type: 8 DH

Lease / Well	Status	Type	Operator	Depth	Dist
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	7.2 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	7.2 mi
McKenzie-Hazard Syndicate	Plugged	DH	McKenzie-Hazard Syndicate	—	7.4 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	8.6 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	8.8 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	9.5 mi
Capell Oil Corp.	Idle	DH	Capell Oil Corp.	—	9.7 mi
Cordelia Oil & Gas Co.	Plugged	DH	Cordelia Oil & Gas Co.	—	12 mi

AI Assessment — Well Proximity

Well proximity is not a planning concern for this site. The nearest well is a plugged Mobil Oil Corporation well 7.2 miles away, which is well beyond any actionable review distance. No active producers, injection wells, or other well types exist within distances that would trigger environmental or surface rights assessments.

Source: California public well records. Active = currently producing or permitted. Plugged = abandoned/decommissioned. Data updated nightly.

200 **10** **108**
total in search area within 5 mi within 25 mi

Site Name	Status	Commodities	Distance
Roderick Sand and Gravel Pit	Occurrence	Sand/Gravel	1.6 mi
Basalt Rock Co	Producer	Clay	2 mi
Cicero Pumice Quarry	Occurrence	PUM	2.9 mi
Mount George King Pumice Deposit.	Occurrence	PUM	2.9 mi
Basalt Rock Co.	Occurrence	DIT	3.1 mi
Mt George Pumice Deposit	Producer	PUM	3.1 mi
Wilson Pumice Quarry	Occurrence	PUM	3.7 mi
Unknown	Occurrence	Mercury	4.4 mi
Silva Pumice Quarry	Occurrence	PUM	4.7 mi
Silva Pumice Quarry	Producer	PUM	5 mi
Walker Pumice Quarry	Occurrence	PUM	5.1 mi
Walker Pumice Quarry	Producer	PUM	5.1 mi
Mountain	Occurrence	Mercury	5.4 mi
Unknown	Occurrence	Mercury	7.6 mi
Unknown	Occurrence	Stone/Crushed	7.8 mi
Unnamed Location	Past Producer	Sand/Gravel	8.5 mi
Unnamed Location	Past Producer	Sand/Gravel	8.5 mi
Unnamed Prospect	Occurrence	Chromium	8.8 mi
Unnamed Location	Occurrence	Chromium	9 mi
Unknown	Occurrence	Chromium	9.1 mi
Bella Oak	Past Producer	Mercury	9.4 mi
Unknown	Occurrence	MG	9.7 mi
Serres Pit	Past Producer	Sand/Gravel	10.4 mi
Unnamed Location	Unknown	Sand/Gravel	10.5 mi
Conn Valley	Occurrence	Manganese	10.6 mi

Showing 25 of 200 deposits. Full dataset via USGS MRDS.

AI Assessment — Mining Context

Site Mineral Resource Review Three active producers operate within 5 miles of the site. Basalt Rock Co (2 mi) and Mt George Pumice Deposit (3.1 mi) are the closest; both will generate heavy truck traffic during operating hours and associated noise from processing operations. Silva Pumice Quarry (5 mi) is also active and may contribute dust and equipment noise depending on operational intensity. No historical deposits are located within 1 mile of the site.

Source: USGS Mineral Resources Data System (MRDS). Status: Producer = active/recent; Past Producer = historical; Prospect = explored not mined; Occurrence = mineralization noted.

ENDANGERED & THREATENED SPECIES - USFWS Critical Habitat within 25mi

6 **6** **12**
Endangered Threatened Total w/ Critical Habitat

6 Endangered species with designated critical habitat. ESA Section 7 consultation may be required for federal nexus projects.

COMMON NAME	SCIENTIFIC NAME	STATUS
California tiger Salamander	<i>Ambystoma californiense</i>	Endangered
Conservancy fairy shrimp	<i>Branchinecta conservatio</i>	Endangered
Contra Costa goldfields	<i>Lasthenia conjugens</i>	Endangered
Soft bird's-beak	<i>Cordylanthus mollis</i> ssp. <i>mollis</i>	Endangered
Suisun thistle	<i>Cirsium hydrophilum</i> var. <i>hydrophilum</i>	Endangered
Vernal pool tadpole shrimp	<i>Lepidurus packardii</i>	Endangered
Alameda whipsnake (=striped racer)	<i>Masticophis lateralis euryxanthus</i>	Threatened
California red-legged frog	<i>Rana draytonii</i>	Threatened
Delta smelt	<i>Hypomesus transpacificus</i>	Threatened
Northern spotted owl	<i>Strix occidentalis caurina</i>	Threatened
Vernal pool fairy shrimp	<i>Branchinecta lynchi</i>	Threatened
western snowy plover	<i>Charadrius nivosus nivosus</i>	Threatened

AI SPECIES & ESA ASSESSMENT - CLAUDE HAIKU

ESA Implications Assessment – 38.3459, -122.3047 **Section 7 Consultation Triggers:** Federal permits, funding, or approvals (water rights, wetland authorizations under Section 404, USDA programs) would trigger mandatory Section 7 consultation given the 12 species with critical habitat within 25 miles and six listed species present. **Most Planning-Relevant Species:** California tiger salamander, vernal pool fairy shrimp, and California red-legged frog warrant immediate attention due to their association with seasonal wetlands and vernal pools —common development constraints in this region—while Delta smelt implicates water supply and conveyance operations. **Recommended Next Step:** Commission a Phase 1 biological resources assessment including wetland delineation, habitat mapping, and species-specific surveys (particularly during breeding seasons) to establish baseline conditions and identify project-specific avoidance/minimization strategies before design finalization. **For planning purposes only. Consult a licensed environmental biologist for formal ESA compliance.**

Source: USFWS Critical Habitat Final Designated Areas - FWS Open Data - Critical habitat does not equal species range

NATIVE PLANTS - iNaturalist Research-Grade within 10mi

57 5051 2026-07-04
Unique Species Total Observations Most Recent

COMMON NAME	SCIENTIFIC NAME	LAST OBS
American black nightshade	<i>Solanum americanum</i>	2026-05-29
Bearded Clover	<i>Trifolium barbigerum</i>	2026-05-15
bird's foot cliffbrake	<i>Pellaea mucronata</i>	2026-04-28
Blow Wives	<i>Achyrachaena mollis</i>	2026-05-20
blue oak	<i>Quercus douglasii</i>	2026-05-09
Bolander's Phacelia	<i>Phacelia bolanderi</i>	2026-04-27
box elder	<i>Acer negundo</i>	2026-05-18
Braun's Giant Horsetail	<i>Equisetum telmateia braunii</i>	2026-06-22
California bay	<i>Umbellularia californica</i>	2026-06-24
California beeplant	<i>Scrophularia californica</i>	2026-05-20
California black oak	<i>Quercus kelloggii</i>	2026-05-15
California buckeye	<i>Aesculus californica</i>	2026-06-02
California Dutchman's Pipe	<i>Aristolochia californica</i>	2026-05-26
California milkwort	<i>Rhinotropis californica</i>	2026-05-15
California mugwort	<i>Artemisia douglasiana</i>	2026-06-02
California poppy	<i>Eschscholzia californica</i>	2026-06-24
California Wild Rose	<i>Rosa californica</i>	2026-05-09
Canyon liveforever	<i>Dudleya cymosa</i>	2026-05-30
-	<i>Castilleja densiflora densiflora</i>	2026-05-09
clay mariposa lily	<i>Calochortus argillosus</i>	2026-05-15
coast live oak	<i>Quercus agrifolia</i>	2026-05-13
coastal woodfern	<i>Dryopteris arguta</i>	2026-05-20
common selfheal	<i>Prunella vulgaris</i>	2026-06-22
Diogenes' lantern	<i>Calochortus amabilis</i>	2026-05-15
distant phacelia	<i>Phacelia distans</i>	2026-05-15
-	<i>Downingia concolor concolor</i>	2026-05-15
Dwarf Brodiaea	<i>Brodiaea terrestris</i>	2026-05-15
Elegant Clarkia	<i>Clarkia unguiculata</i>	2026-05-16
goldback fern	<i>Pentagramma triangularis</i>	2026-05-20
Ithuriel's Spear	<i>Triteleia laxa</i>	2026-06-15

COMMON NAME	SCIENTIFIC NAME	LAST OBS
maroonspot calicoflower	<i>Downingia concolor</i>	2026-05-15
narrowleaf milkweed	<i>Asclepias fascicularis</i>	2026-06-25
narrowleaf mule-ears	<i>Wyethia angustifolia</i>	2026-05-15
Narrowleaf Willow	<i>Salix exigua</i>	2026-06-24
needleleaf navarretia	<i>Navarretia intertexta</i>	2026-05-15
northern California black walnut	<i>Juglans hindsii</i>	2026-06-24
orange bush monkeyflower	<i>Diplacus aurantiacus</i>	2026-05-29
Oregon Ash	<i>Fraxinus latifolia</i>	2026-06-02
Pacific Bleeding Heart	<i>Dicentra formosa</i>	2026-05-04
Pacific madrone	<i>Arbutus menziesii</i>	2026-07-04
Pacific poison oak	<i>Toxicodendron diversilobum</i>	2026-06-02
pineapple-weed	<i>Matricaria discoidea</i>	2026-05-20
Redwood Sorrel	<i>Oxalis smalliana</i>	2026-05-28
small-headed clover	<i>Trifolium microcephalum</i>	2026-05-15
Superb Mariposa Lily	<i>Calochortus superbus</i>	2026-06-11
thimble clover	<i>Trifolium microdon</i>	2026-05-15
tomcat clover	<i>Trifolium willdenovii</i>	2026-05-15
Toyon	<i>Heteromeles arbutifolia</i>	2026-06-19
trailing blackberry	<i>Rubus ursinus</i>	2026-06-24
Tree Clover	<i>Trifolium ciliolatum</i>	2026-05-15
valley oak	<i>Quercus lobata</i>	2026-06-24
wavy-leafed soap plant	<i>Chlorogalum pomeridianum</i>	2026-05-20
western sycamore	<i>Platanus racemosa</i>	2026-05-17
white brodiaea	<i>Triteleia hyacinthina</i>	2026-05-15
white-tipped clover	<i>Trifolium variegatum</i>	2026-05-15
Winecup Clarkia	<i>Clarkia purpurea</i>	2026-05-15
Yellow Mariposa Lily	<i>Calochortus luteus</i>	2026-06-11

AI PLANT COMMUNITY SUMMARY - CLAUDE HAIKU

Ecological Assessment Summary This site in Sonoma County supports a **mixed oak woodland and chaparral transitional community**, dominated by drought-tolerant species like blue oak, California black oak, and California buckeye alongside native shrubs and herbaceous plants adapted to Mediterranean climates. The presence of moisture-loving species such as Braun's Giant Horsetail and California bay indicates seasonal drainage patterns with intermittent water flow, suggesting the site contains riparian or seep areas despite overall xeric conditions. Soils are likely well-drained in upland areas but retain moisture seasonally in low-lying zones, meaning grading or drainage work could disrupt delicate hydrological balance—particularly for the horsetail populations and associated wetland flora that depend on consistent groundwater or seasonal saturation. Any site preparation should preserve existing drainage features and avoid compaction in moist microsites to maintain the ecological integrity of this biodiverse native plant community. Consult a licensed botanist for site-specific vegetation assessment.

Source: iNaturalist research-grade observations - Native species filter applied

FIELD HAZARDS - Venomous & Toxic Species within 25mi

635	154	91	1439
Venomous Snakes	Venomous Spiders	Stinging Insects	Toxic Plants
VENOMOUS SNAKES			
SPECIES		OBSERVATIONS	
Crotalus oreganus oreganus		635 obs	
VENOMOUS SPIDERS			
SPECIES		OBSERVATIONS	
Latrodectus hesperus		154 obs	
STINGING INSECTS			
SPECIES		OBSERVATIONS	
Dasymutilla aureola		91 obs	
Dasymutilla californica		91 obs	
Dasymutilla sackenii		91 obs	
TOXIC PLANTS			
SPECIES		OBSERVATIONS	
Toxicodendron diversilobum		1100 obs	
Datura wrightii		57 obs	

SPECIES	OBSERVATIONS
<i>Datura stramonium</i>	57 obs
<i>Datura ferox</i>	57 obs
<i>Conium maculatum</i>	199 obs
<i>Urtica gracilis</i>	83 obs
<i>Urtica gracilis holosericea</i>	83 obs

AI FIELD SAFETY BRIEFING - CLAUDE HAIKU

Field Safety Brief – Wine Country Survey Area ****Highest-Priority Hazards:**** Poison oak (*Toxicodendron diversilobum*) is your dominant concern with 1,439 observations—expect it throughout this region in virtually all habitats. Northern Pacific rattlesnakes (*Crotalus oreganus oreganus*) are common and pose genuine risk; black widow spiders are present but less frequent. *Datura* species (jimsonweed) and poison hemlock are toxic if contacted or ingested. ****Practical Avoidance:**** Wear long pants, long sleeves, and closed boots; wash clothing separately after fieldwork to remove poison oak oils. Stay on cleared survey routes and avoid brushy areas where rattlesnakes shelter; watch for coiled snakes and listen for rattles, especially in tall grass. Never touch or handle unfamiliar plants, particularly white-flowered plants and those with musty odors. Carry a first-aid kit with tweezers (for splinters/spines) and know the location of the nearest hospital. ****Seasonal Note:**** Spring through fall presents highest snake and insect activity; poison oak is irritating year-round but oils concentrate in warmer months. *Datura* blooms summer through fall. Always verify current conditions locally before fieldwork.

Source: iNaturalist research-grade observations - Always verify locally before fieldwork

FCC MICROWAVE PATHS — Licensed beam paths crossing within 1320ft of site

3	545ft	3
Beam Paths	Closest Beam	Fixed P2P

TX CALL	RX CALL	TYPE	PATH MI	CLOSEST
WBO61	WBO59	Fixed Point-to-Point	—	545ft
KYH37	WQAT627	Fixed Point-to-Point	—	927ft
WRCR329	WQAT627	Fixed Point-to-Point	—	1289ft

Source: FCC ULS microwave license database — PostGIS spatial query — Paths within 1320ft (0.25mi) of site — Towers may be 50+ miles away — FCC Part 101 licensed paths

FAA AIRWAYS — ATS Routes crossing site area

26	13	2	11
Total Airways	Jet Routes	Victor Airways	Other Routes

AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
J1	Jet Route	High	18,000 ft	—	303.8°	121.4°
J110	Jet Route	High	18,000 ft	—	137.9°	318.2°
J126	Jet Route	High	18,000 ft	—	325.5°	144.0°
J143	Jet Route	High	18,000 ft	—	325.0°	145.8°
J3	Jet Route	High	18,000 ft	—	342.8°	161.8°
J32	Jet Route	High	18,000 ft	—	19.2°	199.6°
J501	Jet Route	High	18,000 ft	—	317.1°	135.4°
J58	Jet Route	High	18,000 ft	—	65.3°	245.9°
J65	Jet Route	High	18,000 ft	—	304.1°	121.1°
J80	Jet Route	High	18,000 ft	—	65.3°	245.9°
J84	Jet Route	High	18,000 ft	—	50.8°	231.0°
J88	Jet Route	High	18,000 ft	—	308.0°	127.3°
J94	Jet Route	High	18,000 ft	—	65.3°	245.9°
Q1	Q-Route	High	—	—	334.5°	154.1°
Q120	Q-Route	High	—	—	38.1°	219.8°
Q122	Q-Route	High	—	—	37.8°	219.6°
Q124	Q-Route	High	—	—	37.8°	219.6°
Q138	Q-Route	High	—	—	47.7°	229.0°
Q3	Q-Route	High	—	—	164.0°	344.6°
Q5	Q-Route	High	—	—	161.4°	341.7°
T257	T-Route	Low	—	—	321.1°	140.5°
T259	T-Route	Low	—	—	41.6°	221.8°
T261	T-Route	Low	—	—	31.4°	211.7°
T263	T-Route	Low	—	—	308.8°	128.2°
V107	Victor Airway	Low	7,000 ft	—	295.4°	114.2°
V108	Victor Airway	Low	4,500 ft	—	117.7°	296.8°

RAILROAD PROXIMITY — FRA North American Rail Network

8
Rail Lines

OWNER / OPERATOR	TYPE	TRACKS	PASSENGER	STRACNET	SUBDIVISION
CFNR	Other	1	No	No	—
CFNR	Yard	1	No	No	—
CFNR	Mainline	1	No	No	VALLEJO
SMRT	Mainline	1	No	No	BRAZOS JCT.
NVRR	Mainline	1	No	No	MARTINEZ
USG	Other	1	No	No	—
CFNR	S	1	No	No	—
NVRR	Yard	1	No	No	—

Source: FRA North American Rail Network (NARN) — Updated July 2025 — STRACNET = Strategic Rail Corridor Network (DOD)

WILDLIFE & NATURAL HISTORY - iNaturalist within 25mi

200 90 65 5 14 6
Total Observations Total Species Birds Mammals Reptiles Amphibians

BIRDS (65 species)

COMMON NAME	Scientific Name	OBS
Wild Turkey	<i>Meleagris gallopavo</i>	10
Snowy Egret	<i>Egretta thula</i>	7
California Quail	<i>Callipepla californica</i>	5
Western Bluebird	<i>Sialia mexicana</i>	5
California Towhee	<i>Melospiza crissalis</i>	5
Great Egret	<i>Ardea alba</i>	4
Red-tailed Hawk	<i>Buteo jamaicensis</i>	4
Black Phoebe	<i>Sayornis nigricans</i>	4
Cooper's Hawk	<i>Astur cooperii</i>	3
Ash-throated Flycatcher	<i>Myiarchus cinerascens</i>	3
Western Screech-Owl	<i>Megascops kennicottii</i>	3
Turkey Vulture	<i>Cathartes aura</i>	3
Mallard	<i>Anas platyrhynchos</i>	3
Bushtit	<i>Psaltiriparus minimus</i>	3
Canada Goose	<i>Branta canadensis</i>	3

+ 50 additional species

MAMMALS (5 species)

COMMON NAME	Scientific Name	OBS
Black-tailed Jackrabbit	<i>Lepus californicus</i>	4
Coyote	<i>Canis latrans</i>	2
Mule Deer	<i>Odocoileus hemionus</i>	2
Eastern Fox Squirrel	<i>Sciurus niger</i>	1
Striped Skunk	<i>Mephitis mephitis</i>	1

REPTILES (14 species)

COMMON NAME	Scientific Name	OBS
Western Fence Lizard	<i>Sceloporus occidentalis</i>	10
Pacific Gopher Snake	<i>Pituophis catenifer catenifer</i>	7
Northern Pacific Rattlesnake	<i>Crotalus oreganus oreganus</i>	6
California King Snake	<i>Lampropeltis californiae</i>	3
Pond Slider	<i>Trachemys scripta</i>	2
Western Skink	<i>Plestiodon skiltonianus</i>	2
Ring-necked Snake	<i>Diadophis punctatus</i>	1
Common Garter Snake	<i>Thamnophis sirtalis</i>	1

COMMON NAME	Scientific Name	OBS
Western Yellow-bellied Racer	Coluber constrictor mormon	1
Pacific Ringneck Snake	Diadophis punctatus amabilis	1
Northwestern Pond Turtle	Actinemys marmorata	1
Common Sharp-tailed Snake	Contia tenuis	1
North American Racer	Coluber constrictor	1
Southern Alligator Lizard	Elgaria multicarinata	1

AMPHIBIANS (6 species)

COMMON NAME	Scientific Name	OBS
Pacific chorus frog	Pseudacris regilla	5
American Bullfrog	Lithobates catesbeianus	3
Foothill Yellow-legged Frog	Rana boylei	3
Western Toad	Anaxyrus boreas	2
Rough-skinned Newt	Taricha granulosa	1
California Giant Salamander	Dicamptodon ensatus	1

AI WILDLIFE & ECOLOGICAL ASSESSMENT - CLAUDE HAIKU

Ecological Character Assessment **Location: 38.3459, -122.3047 (Northern California)** This site supports a diverse wildlife community characteristic of oak woodland and grassland habitat with riparian or wetland areas, as evidenced by the co-occurrence of upland species (California Quail, Western Bluebird, California Towhee) and wetland-dependent taxa (egrets, chorus frogs, pond turtles). Several species of regulatory concern are present, including Foothill Yellow-legged Frog and Northwestern Pond Turtle (sensitive to habitat degradation), along with predatory raptids (Red-tailed and Cooper's Hawks) indicating intact food webs. The robust reptile diversity and presence of multiple native amphibian species suggest relatively good habitat connectivity and water availability, though the occurrence of non-native species (American Bullfrog, Eastern Fox Squirrel, Pond Slider) indicates some prior or ongoing anthropogenic disturbance. Overall, this site represents moderately intact natural habitat with capacity to support sensitive species but warrants careful evaluation of potential impacts to riparian corridors, seasonal wetlands, and upland vegetation communities during any proposed site development. *For planning purposes only. Consult a qualified biologist for site-specific biological assessments.*

Source: iNaturalist research-grade observations — Data reflects community science observations and may not represent complete species inventory

✳️ PASSIVE + ACTIVE SOLAR PLANNING — Lat 38.35°

1 • AVERAGE SUN HOURS

Period	Peak Hrs/Day	Notes
Annual average	6.6 hrs	Overall solar resource
Summer average	9 hrs	June–August
Winter average	3.9 hrs	December–February
Worst month (Dec)	3.2 hrs	Size battery storage to this

2 • IDEAL SOLAR ARRAY PITCH

Optimal tilt = 38.3° (latitude) → 9/12 pitch

Pitch	Angle	% of Optimal	Notes
4/12	18.4°	97%	good
5/12	22.6°	97.6%	good
6/12	26.6°	98.2%	↔ excellent
7/12	30.3°	98.8%	↔ excellent
8/12	33.7°	99.3%	↔ excellent
9/12	36.9°	99.8%	↔ annual optimal
10/12	39.8°	99.8%	↔ excellent
12/12	45°	99%	↔ excellent

Going from 5/12 to 6/12 at framing costs ~\$800–1,200 one time. On a 10kW system that's +723 kWh/yr (~\$108 value/yr). Over 25yr panel life: ~\$2710 total gain.

3 • SOLAR HEAT GAIN — BTU/HR PER SQ FT

Use these values as inputs for eQUEST, EnergyPlus, Manual J, or HERS rating

Surface	Winter	Summer	Design Note
South glass (SHGC 0.4)	51	31	Minimize south glass
South glass (SHGC 0.6)	76	46	Clear/uncoated glass
North glass (any)	10	13	Negligible — diffuse only
East / West glass	27	104	⚠ Minimize — afternoon load
Roof — dark (absorb 0.95)	236	301	Highest load surface
Roof — light (absorb 0.35)	87	111	Cool roof — significant reduction
Roof — metal (absorb 0.25)	62	79	Best roof option for heat
South wall — dark	94	69	
South wall — light	31	23	Light color = major savings
Hardscape — asphalt	181	301	⚠ Heat island — avoid near structure
Hardscape — concrete	124	206	Also reflects onto adjacent walls
Hardscape — light pave	67	111	Recommended within 50ft
Pool surface	23	38	Low absorb + evaporative cooling

Concrete reflection onto south wall: ~4 BTU/hr·ft² additional load per 100sf of adjacent hardscape.
Pool specular reflection: angle-dependent — east/west pools can cause morning/afternoon glare into glass.

4 • ROOF PITCH SOLAR LOAD — BTU/HR PER 100 SQ FT (SUMMER)

Pitch	Dark Roof	Light Roof	Reduction vs Flat
0/12	301 BTU	111 BTU	baseline (worst)
4/12	241 BTU	89 BTU	-20% dark / -20% light
6/12	196 BTU	72 BTU	-35% dark / -35% light
8/12	166 BTU	61 BTU	-45% dark / -45% light
12/12	135 BTU	50 BTU	-55% dark / -55% light

Scale: multiply BTU × (your roof sf ÷ 100). Pitch + light color = biggest single cooling move.

5 • THERMAL TIME LAG — WALL ASSEMBLY

Peak outdoor temp ~15:00. Heat arrives at interior = peak + lag hours.

Assembly	Lag	Peak Interior Load	Note
2x4 frame (R-15)	0.6 hrs	~15:36 PM	R-value only — no mass benefit
2x6 frame (R-23)	0.9 hrs	~15:54 PM	R-value only — no mass benefit
2x8 frame (R-30)	1.2 hrs	~16:12 PM	R-value only — no mass benefit
2x10 frame (R-37)	1.5 hrs	~16:30 PM	good
2x12 frame (R-44)	1.8 hrs	~16:48 PM	good
ICF 6" core	7 hrs	~22:00 PM	✓ shifts past sunset — natural vent window
8" CMU block	6 hrs	~21:00 PM	✓ shifts past sunset — natural vent window
12" concrete	8.5 hrs	~23:30 PM	✓ shifts past sunset — natural vent window

In 3C Marine: ICF or CMU shifts peak load past sunset. Open windows, flush heat naturally. AC load drops significantly.

6 • TREE SHADOW PLANNING — TREE DUE SOUTH, 100FT FROM STRUCTURE

Tree Ht	Shadow Reaches	Summer hrs/day	Equinox hrs/day	Winter hrs/day	WUI note
10ft	All year	1.1	1	1.2	OK
20ft	All year	2.1	1.9	2.4	OK
30ft	All year	3.1	2.9	3.7	Zone 2 check
40ft	All year	4	3.8	5.2	△ verify
50ft	All year	4.8	4.6	7.3	△ verify
60ft	All year	5.6	5.5	9.3	✗ Zone 1

Deciduous tree: winter solar penetration ~85% (leaves off) — best of both worlds in hot-dry climates.

Evergreen tree: winter penetration ~15% — good windbreak, poor solar strategy south of structure.

△ WUI zones: tree placement must be coordinated with defensible space requirements. Verify with local fire authority — WUI boundaries vary by jurisdiction.

7 • ATTIC VENTING — PER 100 SQ FT OF ROOF AREA

Standard	Unbalanced (1/150)	Balanced† (1/300)	Notes
Code minimum	8 sq in	4 sq in	Moisture protection only
Thermal comfort	24 sq in	12 sq in	Keeps attic within 20°F of outdoor
WUI mesh 1/16"	60 sq in	30 sq in	2.5× compensation — mesh clogs
WUI intumescent	24 sq in	12 sq in	Self-closing — no compensation needed

Pitch factor: 0/12×1.0 · 4/12×0.80 · 6/12×0.65 · 8/12×0.55 · 12/12×0.45

Color factor: Dark×1.0 · Medium×0.75 · Light×0.55 · Metal×0.45

Scale: multiply sq in × (your roof sf ÷ 100) × pitch factor × color factor

†**Balanced** = ≥40% NFA within top third of roof height (ridge) + ≥40% within bottom third (soffit)

WUI conflict: Soffit vents are ember collection points. In high WUI zones — eliminate soffit vents, use ridge-only (unbalanced/1/150). In low WUI — balanced system preferred.

△ Verify WUI zone with local fire authority — boundaries vary significantly by jurisdiction.

🏠 AI SOLAR & PASSIVE DESIGN SUMMARY — HOMEOWNER GUIDE

Your Home's Solar & Energy Plan Your location gets about 6.6 hours of strong sunshine per day on average—think of it as having a moderately generous solar battery charging window year-round, though it shrinks to just over 3 hours in winter and expands to 9 hours in summer. A dark roof acts like a heat sponge, absorbing over 300 BTUs per hour for every square foot on hot days, which is why switching to a light-colored roof can feel like installing free air conditioning; fortunately, your roof's ideal tilt of about 9:12 pitch (which is pretty standard) naturally angles toward the sun for solid panel efficiency. Your west-facing windows are the real trouble spot during summer afternoons—they're letting in heat equivalent to 104 BTUs per hour per square foot, which explains why west-facing rooms can feel like ovens in late afternoon, so strategic shading or reflective glass there pays dividends. Here's the smart part: a 40-foot deciduous tree planted on your south side will block unwanted summer heat for about 4 hours while still letting winter sun through for 5+ hours (since the tree's bare in winter), essentially giving you free seasonal temperature control. Thermal mass—basically, heavy materials like concrete or masonry that absorb heat slowly and release it slowly—acts like your home's battery, smoothing out temperature swings throughout the day. These numbers are planning estimates—your architect or energy modeler can plug them directly into their software for a full analysis.

Planning estimates only — clear-sky model, standard assumptions. Values formatted for eQUEST / EnergyPlus / Manual J input. Verify with licensed architect, energy modeler, or solar installer before specifying.

Technical Brief — for architects, engineers, and surveyors

SITE INTELLIGENCE BRIEFING ****38.34586°N, 122.30466°W | Napa County, California****

****Structural Design Drivers & Climate Envelope****

This site falls within IECC Climate Zone 3C (Marine) with an annual freezing index of 0°F-days, indicating no sustained winter frost threat and minimal thermal cycling stress on exposed materials. Wind, snow, and seismic design parameters are not populated in the available dataset, creating a critical gap: absent explicit seismic demand data, assume California Building Standards mandate per California Building Code requirements for Napa County's seismic environment. The marine climate classification (Zone 3C) governs moisture management and material durability; coastal-influenced weather patterns suggest attention to vapor drive and corrosion risk, though the site's inland elevation and distance from the coast require site-specific wind speed verification. Do not proceed with structural framing without obtaining site-specific seismic design parameters and confirmed wind speed exposure for the immediate location.

****Geotechnical Foundation Design****

The site is underlain by Coombs gravelly loam, 2–5 percent slopes, classified as Hydrologic Group C with well-drained characteristics. Group C soils indicate moderate infiltration rates and moderate permeability; the well-drained designation supports conventional shallow foundations without extensive dewatering provisions. The 2–5 percent slope suggests minor topographic relief but does not preclude level building pads with standard cut-and-fill; however, slope stability analysis is warranted given the fine-grained loam fraction and potential for shallow seepage along the slope face. Frost depth is not provided—contact the local building authority (Napa County) for the frost-protected foundation depth requirement, as California County codes vary. The soil texture field lacks completion; obtain a complete geotechnical phase I (boring log with Atterberg limits, bearing capacity, and settlement estimates) to confirm foundation type, depth, and any need for ground improvement.

****Hazard Assessment****

No Quaternary faults are identified within the search radius, and no subsidence basin has been mapped at this location. This does not eliminate seismic risk; Napa County is seismically active, and the absence of nearby mapped faults does not exclude moderate-to-high ground motion. Wildfire data were not provided; however, Napa County's fire history warrants consultation with Cal Fire Hazard Severity Zones. No oil and gas activity is present at the site; the nearest well (Mobil Oil Corporation, 7.2 miles distant) is plugged and not producing, eliminating subsurface resource extraction risk.

****Mining & Site Context****

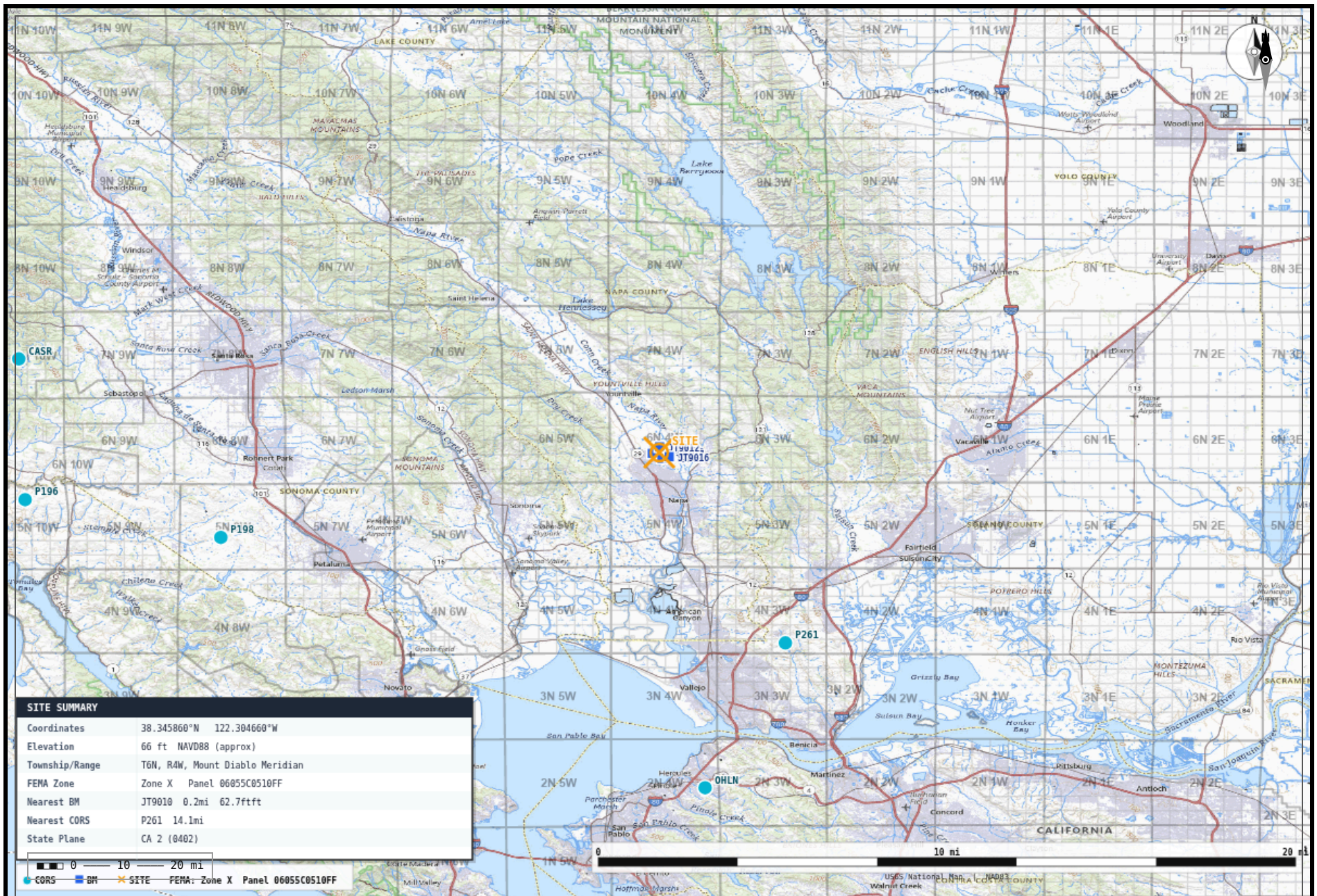
Three active mining operations lie within 5 miles: Basalt Rock Co (clay, 2 miles), Mt George Pumice Deposit (3.1 miles), and Silva Pumice Quarry (5 miles). While none are immediately proximate, cumulative 200 MRDS deposits within ~35 miles reflects the region's mineral-rich geology. Mining activity can affect groundwater quality and subsurface stability; confirm whether any active operations trigger vibration, noise, or blast-shock requirements and verify groundwater chemistry if the site draws potable or process water. The marine climate and well-drained soils support conventional construction, but the regional mining context warrants Phase II environmental assessment.

****Bottom-Line Directive****

Obtain site-specific seismic design parameters, confirmed frost depth requirement from Napa County Building Authority, and complete Phase I geotechnical report with bearing capacity and settlement analysis before finalizing foundation design.


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Planning aid only — not a substitute for licensed professional review, geotechnical investigation, or code compliance verification.

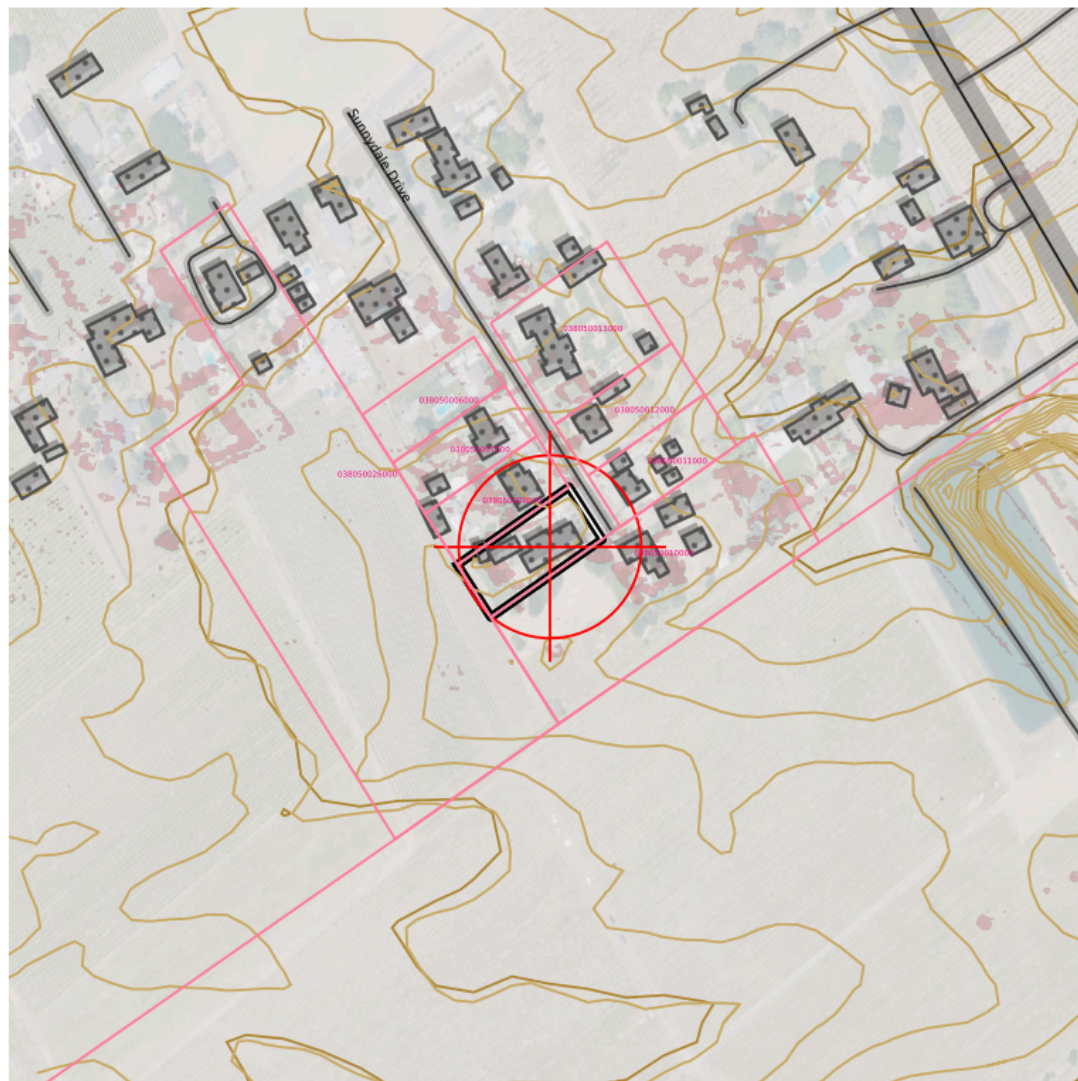


TOPOSCOUT
FIELD MAP

TopoScout Field Map
Lat: 38.34586 Lng: -122.30466
38.345860°N 122.304660°W | Radius: 30 mi

Generated: July 13, 2026
Base: USGS National Map
Generated by TopoScout - outkastdesigns.com

TOPOSCOUT SITE PLAN PREVIEW | \$19 UPGRADE FOR AUTOCAD DXF FILE



INCLUDED LAYERS



0 — 400 ft State Plane · NAD83 · US Feet

- Contours (Index/Major/Minor/1ft)
- PLSS Section/Township/QQ
- Roads Major/Local/Trail + ROW
- Buildings
- Hydro
- Power Lines
- Pipeline
- Railroad
- Benchmarks + Labels
- County Boundary
- OTLS Grid
- Site Info
- Elevation Labels
- 3D Polylines (all geometry)

✦ FIELD RESOURCE LINKS

PRE-MISSION PLANNING

Trimble GNSS Planning Online –

<https://trimblegnssplanningonline.com/>

Satellite visibility, DOP charts, sky plots

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Drone airspace authorization, TFRs, restricted areas

Aloft (Kittyhawk) – <https://www.aloft.ai/>

Field-friendly drone airspace map and LAANC authorization

NOAA Space Weather — Live Kp –

<https://www.swpc.noaa.gov/products/planetary-k-index>

Real-time Kp index, geomagnetic storm alerts

POST-PROCESSING

NGS OPUS – <https://geodesy.noaa.gov/OPUS/index.jsp>

Online Positioning User Service – static GNSS processing

NGS OPUS-RS – <https://geodesy.noaa.gov/OPUSI/index.jsp>

Rapid static processing – sessions as short as 15 minutes

CORS Data Download – <https://geodesy.noaa.gov/CORS/cors-data.shtml>

Raw RINEX observation files from CORS stations

FEMA & FLOOD

FEMA Flood Map Service Center –

<https://msc.fema.gov/portal/home>

FIRM panels, flood zone lookup by address or coordinates

FEMA Elevation Certificate – [https://www.fema.gov/flood-](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

[insurance/work-with-nfip/elevation-certificate](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

Current EC form, instructions, LOMA/LOMR submission guidance

BASE STATION & NETWORK

NOAA CORS Network – <https://geodesy.noaa.gov/CORS/cors-data.shtml>

Continuously operating reference stations, data download

NGS CORS Station Map –

<https://geodesy.noaa.gov/CORS/GoogleMap/CORSmap.shtml>

Interactive map of all CORS stations with status

IGS Station Monitor – <https://www.igs.org/network>

International GNSS Service – global base station health

REFERENCE & CALCULATIONS

NGS Datasheets – <https://www.ngs.noaa.gov/cgi-bin/datasheet.prl>

Benchmarks, control points, passive monuments

NGS Coordinate Conversion (NCAT) –

<https://geodesy.noaa.gov/NCAT/>

NADCON5, datum transforms, state plane conversions

NOAA Magnetic Declination –

<https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml>

Declination calculator for any location and date

NGS Geoid Models – [https://geodesy.noaa.gov/GEOID/](https://geodesy.noaa.gov/GEOID/GEOID18)

[GEOID18](https://geodesy.noaa.gov/GEOID/GEOID18) and current model downloads, GEOID calculator

DRONE OPERATIONS

FAA Part 107 — Commercial Rules –

https://www.faa.gov/uas/commercial_operators

Full Part 107 regulatory summary for commercial UAS operations

FAA DroneZone – <https://faadronezone.faa.gov/>

Drone registration, operator certificates, waiver management

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Pre-flight airspace check, TFRs, LAANC authorization

FAA Part 107 Waivers –

https://www.faa.gov/uas/commercial_operators/part_107_waivers

Waiver applications for operations beyond standard 107 rules

Aloft — Field Airspace Map – <https://www.aloft.ai/>

Mobile-friendly real-time airspace, LAANC, flight logging

AIR QUALITY & WILDFIRE SMOKE

[AirNow Fire & Smoke Map](https://www.fire.airnow.gov)

<https://www.fire.airnow.gov>

Current fire locations, smoke plumes, PM2.5 readings by location

[AirNow Interactive Map](https://gispub.epa.gov/airnow)

<https://gispub.epa.gov/airnow>

Real-time AQI monitors, hourly NowCast, 48-hr forecast overlays

[NOAA Smoke Forecast](https://airquality.weather.gov)

<https://airquality.weather.gov>

48-hr smoke concentration forecast, HRRR-Smoke model

